Unit 4: Reproduction

Complete Study Notes – Grade 9 Biology

4.1 Introduction to Reproduction

Reproduction is one of the unique characteristics of life. The ability of organisms to reproduce to form their own kind is the one characteristic that best distinguishes living things from non■living matter.

Two modes of reproduction are recognised: **asexual reproduction** and **sexual reproduction**.

In **asexual reproduction**, there is only one parent, and no special reproductive organs or cells are involved. Each organism is capable of producing identical copies of itself (clones).

In **sexual reproduction**, two parents are usually involved, and special reproductive cells (gametes) are produced. Offspring are genetically different from each other and from the parents.

No organism can live forever, but part of it lives in its offspring. Offspring are produced by the process of reproduction, which ensures the continuation of the species.

4.2 Asexual Reproduction

Asexual reproduction is the production of individuals without gametes (eggs or sperm). It includes a number of distinct processes, all without involving sex or a second parent.

Asexual reproduction appears in bacteria, unicellular eukaryotes, many invertebrates, fungi, and plants. However, asexual reproduction is absent among vertebrates.

The basic forms of asexual reproduction are **fission** (binary and multiple), **budding**, **fragmentation**, **vegetative propagation**, and **spore formation**.

Advantages of Asexual Reproduction:

- No mate is needed.
- No gametes are needed.
- All the good characteristics of the parent are passed on to the offspring.
- Offspring will grow in the same favourable environment as the parent.
- Plants that reproduce asexually usually store large amounts of food that allow survival.

Disadvantages of Asexual Reproduction:

- Little genetic variation is created, so adaptation to a changing environment (evolution) is unlikely.
- If the parent has no resistance to a particular disease, none of the offspring will have resistance.
- Lack of dispersal can lead to competition for nutrients, water, and light.

4.3 Types of Asexual Reproduction

4.3.1 Fission

In fission, the organism divides into two (**binary fission**) or more (**multiple fission**) equal parts.

Binary fission is common among bacteria, algae, and protozoa. The body of the unicellular parent divides by mitosis into two approximately equal parts, each of which grows into an individual similar to the parent. Before a bacterium divides, the bacterial DNA is replicated to produce two identical copies so that each daughter cell receives one copy.

Example: Bacteria can divide every 20–30 minutes. When placed on a solid growth medium in a Petri dish, a bacterial cell and its daughter cells undergo repeated cellular divisions and form a group of genetically identical cells called a **bacterial colony**. This rapid rate of reproduction is why a small number of bacteria can seriously contaminate our food products.

Multiple fission occurs when the nucleus divides repeatedly, and each daughter nucleus breaks away together with a small portion of cytoplasm, resulting in many daughter cells. This is common among some parasitic protozoa, such as malarial parasites.

4.3.2 Fragmentation

Fragmentation is one of the most common modes of asexual reproduction, involving the breakdown of a parent organism into parts that develop into whole organisms.

Fragmentation is observed in fungi, plants, animals, and algae. For example, *Spirogyra* (filamentous green algae) undergoes fragmentation, resulting in many filaments. Each filament grows into a matured filament.

Also, some multicellular animals (e.g., worms) can break into two or more parts, with each fragment capable of becoming a complete individual. Many invertebrates can reproduce asexually by simply breaking into two parts and then regenerating the missing parts.

4.3.3 Budding

In budding, the organism divides into two unequal parts. A bulge forms on the side of the cell, the nucleus divides mitotically, and the bud ultimately detaches itself from the mother cell.

Budding is common in fungal species (e.g., yeast) and invertebrates (e.g., hydra). In yeast, a bud develops on the surface, the nucleus divides, one daughter nucleus migrates into the bud, and the other remains in the parent cell. The bud grows and eventually detaches.

4.3.4 Vegetative Propagation

Vegetative propagation is a method of asexual reproduction in plants where structures with lateral meristems such as roots, stems, buds, and leaves give rise to new self-supporting individuals.

Natural Vegetative Propagation:

- **Stolons (runners) –** Horizontal stems that grow above the ground. At nodes, roots and shoots develop, forming a new plant. Example: strawberry, couch grass.
- **Rhizomes –** Horizontal underground stems. At nodes, buds develop and produce shoots above ground. Example: ginger, couch grass.
- **Comms –** Rounded, fleshy underground stems that store food. Example: gladiolus, taro.
- **Tubers –** Swollen underground stems that store starch. Example: potato (*Solanum tuberosum*).
- **Bulbs –** Underground storage organs with fleshy leaves. Example: onion, garlic.

Artificial Vegetative Propagation:

- **Grafting –** Joining a stem (scion) from one plant onto the root (stock) of another. Combines favourable characteristics.
- **Cutting –** A portion of stem containing nodes and internodes is placed in moist soil to root. Example: hibiscus, crotons.
- **Layering –** A stem attached to the plant is bent and covered with soil. Roots develop at the covered portion.
- **Micropropagation (tissue culture) –** Growing plants from small tissue samples on nutrient medium in the laboratory.

Example: Potato tubers are actually modified stems. They have 'eyes' (buds) that can develop into new plants. If you plant a potato tuber with its eyes, each eye can sprout and grow into a new potato plant.

4.3.5 Parthenogenesis

Parthenogenesis is a form of asexual reproduction in which an **unfertilized egg** develops into an adult organism.

It is common in some insects (e.g., aphids, bees, wasps), some reptiles, and occasionally in fish and amphibians.

Example – Honeybees: In honeybees, the queen lays fertilized eggs that become worker females (diploid), and unfertilized eggs develop into males (drones) through parthenogenesis (haploid). This is a natural mechanism for sex determination.

Because offspring produced by parthenogenesis arise from an egg without sperm, it is sometimes called 'virgin birth'.

4.4 Sexual Reproduction in Humans

Sexual reproduction involves the production of sex cells (**gametes**). It almost always involves two parent organisms.

The gametes are produced in reproductive organs through **meiosis**. Sexual reproduction starts with the union of sperm and an egg in a process called **fertilization**. This can occur either inside (internal fertilization) or outside (external fertilization) the body of the female.

Fertilization results in the formation of a single cell called a **zygote**. The zygote then grows into a new individual by mitosis.

Advantages of Sexual Reproduction:

- • Produces genetic variation among offspring, which may help in adaptation to changing environments.
- • Allows for the elimination of harmful mutations.
- • Increases the chances of survival of the species.

Disadvantages of Sexual Reproduction:

- • Requires two parents (energy and time needed to find a mate).
- • Slower reproductive rate compared to asexual reproduction.
- • Only half of the population (females) produce offspring directly.

4.5 Primary and Secondary Sexual Characteristics

During puberty, humans mature physically and become able to reproduce. Hormones (chemical messengers) control these changes.

Primary Sexual Characteristics: These are the reproductive organs present at birth.

- **Males:** Penis, scrotum, and testes.
- **Females:** Vagina, uterus, and ovaries.

Secondary Sexual Characteristics: These develop during puberty and are not directly involved in reproduction but are essential for reproductive success.

In Males (testosterone-driven):

- • Growth of the male sex organs.
- • Increase in body hair (facial, pubic, chest).
- • Increase in muscle mass and bone growth.
- • Deepening of the voice.
- • Increase in libido.

In Females (estrogen-driven):

- • Growth of breasts and widening of hips.
- • Increase in body hair (underarm, pubic).
- • Accumulation of fat in the breasts, buttocks, and thighs.
- • Onset of the menstrual cycle.
- • Maturation of the ovaries and release of eggs.

Psychological and Social Changes:

Adolescence is a time of significant psychological and social changes. Young people must adjust to rapid physical changes, which can affect self-esteem. Early or late puberty can cause anxiety. It is important to understand that there is considerable variation in the timing of puberty, and most people develop into normal adults.

4.6 Male Reproductive Structures

The male reproductive system includes the **testes**, **epididymis**, **vas deferens**, **accessory glands** (seminal vesicles, prostate gland, bulbourethral glands), and the **penis**.

Testes:

Paired testes are the sites of **sperm production**. Each testis is composed of numerous **seminiferous tubules** where sperm develop. Between the tubules are **Leydig cells** that produce **testosterone** (the male sex hormone). The testes are housed in the **scrotum**, which keeps them at a temperature slightly lower than body temperature (about 2–3°C lower) – essential for viable sperm production.

Sperm Ducts:

Sperm travel from the seminiferous tubules to the **epididymis** where they mature and are stored. From the epididymis, they pass through the **vas deferens** (sperm duct) to the **ejaculatory duct**, which joins the **urethra**. The urethra carries both sperm and urine through the penis.

Accessory Glands:

• **Seminal vesicles –** secrete a fructose-rich fluid that provides energy for sperm.

• **Prostate gland –** produces alkaline secretions that help neutralise acidity in the female reproductive tract.

• **Bulbourethral glands –** secrete a lubricating mucus that helps clear the urethra of residual urine.

Semen is a mixture of sperm and secretions from the accessory glands. Sperm constitute less than 5% of semen volume.

Sperm Structure:

• • **Head:** Contains the nucleus with genetic material and an acrosome (enzyme-rich cap) that helps penetrate the egg.

• • **Midpiece:** Contains many mitochondria that provide energy for flagellar movement.

• • **Tail (flagellum):** Allows the sperm to swim toward the egg.

4.7 Female Reproductive Structures

The female reproductive system includes the **ovaries**, **oviducts** (fallopian tubes), **uterus**, **cervix**, and **vagina**.

Ovaries:

A human female's gonads – her ovaries – lie deep inside her pelvic cavity. They produce and release eggs (ova) and secrete the sex hormones **estrogen** and **progesterone**. Estrogen triggers the development of female secondary sexual characteristics, and progesterone thickens the uterine lining in preparation for pregnancy.

At birth, a female has about 400,000 eggs. Only about 300–400 will mature and be released during her reproductive lifetime (about 30–35 years).

Oviducts (Fallopian Tubes):

Hollow tubes connecting the ovaries to the uterus. They are lined with cilia that help move the egg toward the uterus. Fertilization usually occurs in the oviduct.

Uterus (Womb):

A hollow, pear-shaped organ with thick muscular walls and a specialized lining called the **endometrium**. The uterus houses and nourishes the developing embryo.

Cervix:

The narrowed lower end of the uterus that opens into the vagina. It produces mucus that changes in consistency during the menstrual cycle.

Vagina:

The organ of intercourse and the birth canal. It extends from the cervix to the body's surface.

External Genitals (Vulva):

Includes the **labia majora** (outer folds), **labia minora** (inner folds), **clitoris** (sensitive erectile tissue), and the **hymen** (a membrane that partially covers the vaginal opening in some females).

4.8 The Menstrual Cycle

The menstrual cycle is the approximately monthly series of changes in the female reproductive system that prepares the body for pregnancy.

The cycle is controlled by hormones: **FSH** (follicle-stimulating hormone), **LH** (luteinizing hormone), **estrogen**, and **progesterone**.

Phases of the Menstrual Cycle (average 28 days):

1. **Menstrual Phase (Day 1–5):** The uterine lining (endometrium) breaks down and is shed through the vagina. This is menstruation (period). Low levels of estrogen and progesterone trigger the shedding.
2. **Follicular Phase (Day 1–13):** FSH from the pituitary stimulates the growth of a follicle in the ovary. The follicle produces estrogen, which causes the endometrium to thicken. Estrogen levels rise, stimulating the release of LH.
3. **Ovulation (Day 14):** A surge in LH triggers the release of a mature egg from the ovary (ovulation). The egg is swept into the oviduct and is available for fertilization.
4. **Luteal Phase (Day 15–28):** After ovulation, the ruptured follicle becomes the **corpus luteum**, which secretes estrogen and progesterone. These hormones further thicken the endometrium and prepare it for implantation. If fertilization does not occur, the corpus luteum degenerates, hormone levels drop, and the endometrium breaks down – starting a new cycle.

Menstrual Hygiene:

Good menstrual hygiene practices help prevent infections and discomfort: change pads/tampons regularly, wash the external genital area with water, wear cotton underwear, and track menstrual cycles to monitor health. Adolescence is a time to learn about these practices.

4.9 Fertilization and Pregnancy

Fertilization:

Fertilization occurs when a sperm cell penetrates the egg cell and the nuclei fuse, forming a **zygote**. This usually happens in the oviduct.

After fertilization, the zygote begins to divide by mitosis and forms a ball of cells called a **morula**, which becomes a **blastocyst**. The blastocyst travels down the oviduct and implants into the endometrium of the uterus – this is called **implantation**.

Once implanted, the embryo develops. After 8 weeks, when all major organs have formed, it is called a **fetus**.

Placenta and Umbilical Cord:

The **placenta** forms from embryonic tissues and attaches to the uterine wall. It allows the exchange of oxygen, nutrients, and waste products between the mother and the fetus. The **umbilical cord** connects the fetus to the placenta.

The fetus is surrounded by the **amnion** (a fluid-filled sac) that protects it from physical damage.

Twins:

- **Identical (monozygotic) twins:** One zygote splits into two embryos. They share the same genetic material.

• **Fraternal (dizygotic) twins:** Two separate eggs are fertilized by two different sperm. They are no more alike than siblings.

4.10 Methods of Birth Control

Birth control (contraception) is the deliberate prevention of pregnancy. Methods can be natural, barrier, hormonal, or surgical.

Natural Methods:

• **Abstinence –** Avoiding sexual intercourse. 100% effective for pregnancy and STI prevention.

• **Fertility awareness (rhythm method) –** Avoiding intercourse during the fertile period. Effectiveness ~75–90% with perfect use.

• **Withdrawal (coitus interruptus) –** The male withdraws before ejaculation. Effectiveness ~80–96% with perfect use.

Barrier Methods:

• **Male condom –** A latex/lambskin sheath worn on the penis. Prevents sperm from entering the vagina. Also protects against STIs. Effectiveness ~98% with perfect use.

• **Female condom –** A polyurethane pouch inserted into the vagina. Effectiveness ~95% with perfect use.

• **Diaphragm –** A rubber dome inserted over the cervix. Used with spermicide. Effectiveness ~95% with perfect use.

• **Spermicides –** Chemicals that kill sperm. Often used with other barrier methods.

Hormonal Methods:

• **Contraceptive pill –** Contains estrogen and progesterone that inhibit ovulation. Taken daily. Effectiveness >99% with correct use.

• **Contraceptive implant –** A small rod inserted under the skin of the arm, releases progesterone over 3 years. Effectiveness >99%.

• **Contraceptive injection –** Progesterone injection given every 8–12 weeks. Effectiveness >99%.

• **Intrauterine device (IUD) –** A T-shaped device inserted into the uterus. Can be hormonal or copper. Effectiveness >99%.

Surgical Methods (Sterilization):

• **Vasectomy (male) –** Cutting and sealing the vas deferens, preventing sperm from entering the semen. Permanent and nearly 100% effective.

• **Tubal ligation (female) –** Cutting, tying, or blocking the oviducts to prevent eggs from reaching the uterus. Permanent and nearly 100% effective.

Emergency Contraception:

• **Emergency contraceptive pill –** Taken within 72 hours of unprotected sex. Prevents or delays ovulation. Not for regular use.

4.11 Sexually Transmitted Infections (STIs)

Sexually transmitted infections (STIs) are infections that are spread mainly through sexual contact.

Common STIs in Ethiopia:

Trichomoniasis: Caused by the protozoan *Trichomonas vaginalis*. Symptoms: yellowish discharge, itching, soreness. Treatable with antibiotics.

Chlamydia: Caused by the bacterium *Chlamydia trachomatis*. Often asymptomatic; can cause painful urination, discharge. If untreated, leads to infertility. Treatable with antibiotics.

Gonorrhea: Caused by bacterium *Neisseria gonorrhoeae*. Symptoms: discharge, painful urination. Can cause sterility. Treated with antibiotics (but drug-resistant strains are increasing).

Syphilis: Caused by bacterium *Treponema pallidum*. Produces painless sores (chancres), then systemic symptoms. Can damage brain, heart if untreated. Treatable with antibiotics.

HIV/AIDS: Caused by the human immunodeficiency virus (HIV). HIV attacks immune cells (T-lymphocytes), leading to AIDS. No cure, but antiviral drugs can manage the infection. Transmitted through blood, semen, vaginal fluids, breast milk.

Modes of Transmission:

• Unprotected sexual intercourse (vaginal, anal, oral) with an infected person.

• Blood transfusion with infected blood.

• Sharing contaminated needles (drug use, tattoos).

• Mother-to-child transmission during pregnancy, childbirth, or breastfeeding.

Prevention of STIs:

• • **Abstinence –** The most effective method.

• • **Use of condoms –** Provide strong protection against many STIs, including HIV.

• • **Mutual monogamy** – Having a single sexual partner who is also monogamous.

• • **Testing and treatment** – Early detection and treatment of STIs reduce transmission.

• • **Vaccination** – Vaccines exist for some STIs (e.g., HPV, hepatitis B).

• • **Avoid sharing needles –** Use sterile needles for injections.

HIV/AIDS – Key Facts:

• HIV attacks the immune system, destroying T-lymphocytes (helper T cells). Over time, this leads to AIDS, where the body can no longer fight off opportunistic infections.

• There is no cure for HIV, but antiretroviral therapy (ART) can suppress the virus and prolong life.

• HIV is not transmitted through casual contact (hugging, sharing food, mosquito bites).

Unit 4 Summary

• **Reproduction** is the biological process by which organisms produce offspring. It can be asexual (one parent, no gametes) or sexual (two parents, gametes).

• **Asexual reproduction** includes fission (binary and multiple), fragmentation, budding, vegetative propagation (natural and artificial), and parthenogenesis.

• **Advantages:** No mate needed, fast, all genes passed on. **Disadvantages:** No genetic variation, vulnerable to environmental changes.

• **Sexual reproduction** involves gametes (sperm and egg) formed by meiosis. Fertilization produces a zygote that develops into an embryo.

• **Primary sexual characteristics** are the reproductive organs (testes, ovaries, etc.). **Secondary sexual characteristics** develop during puberty (facial hair, breasts, voice change) driven by hormones (testosterone, estrogen).

• **Male reproductive system:** Testes (sperm production, testosterone), epididymis (sperm maturation), vas deferens, prostate, seminal vesicles, penis.

• **Female reproductive system:** Ovaries (egg production, estrogen/progesterone), oviducts (fertilization site), uterus (fetal development), cervix, vagina.

• **Menstrual cycle:** Regulated by FSH, LH, estrogen, progesterone. Phases: menstrual, follicular, ovulation, luteal.

• **Fertilization** occurs in the oviduct; the zygote implants in the uterus. The placenta and umbilical cord support the fetus.

• **Birth control** methods include natural (abstinence, rhythm), barrier (condoms, diaphragm), hormonal (pill, implant), and surgical (vasectomy, tubal ligation).

• **Sexually transmitted infections (STIs):** Caused by bacteria, viruses, or protozoa. Examples: HIV, syphilis, gonorrhea, chlamydia, trichomoniasis. Prevention: condoms, abstinence, regular testing.

End of Unit 4 – Reproduction